

Detection of Fine-Grained Tennis Events in Campus Videos via Skeleton-Guided Multimodal Distillation

by

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A CAPSTONE PROJECT SUBMITTED FOR THE DEGREE OF

MASTER OF COMPUTING

in the

GRADUATE DIVISION

of the

NATIONAL UNIVERSITY OF SINGAPORE

2026

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Declaration

I hereby declare that this project report is my original work and it has been written by me in its entirety. I have duly acknowledged all the sources of information which have been used in this report.

This report has also not been submitted for any degree in any university previously.

LI LINGYU
February 11, 2026

To my family and friends

Acknowledgments

I would like to express my sincere gratitude to my supervisor for the invaluable guidance and support throughout this project. The insights and expertise have been instrumental in shaping this research.

I am especially grateful to my parents for their unconditional love, encouragement, and sacrifices throughout my studies; their support and guidance have been indispensable to the completion of this project.

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Abstract

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Recognizing fine-grained sports events with precise temporal localization is essential for sports analytics but remains challenging in amateur settings due to rapid succession, motion blur, and subtle visual differences. While professional models achieve high accuracy, they suffer from significant domain shifts when applied to campus-level footage recorded with monocular cameras. This project presents College-Tennis, an automated system designed to bridge the gap between professional and amateur tennis analytics.

We propose a novel pipeline based on the MD-FED (Multimodal Distillation for Few-Shot Fine-Grained Event Detection) framework. The approach consists of five strategic steps: (1) Initial temporal segmentation and pseudo-labeling using a pre-trained F3Set model; (2) High-resolution skeleton extraction using HRNet-w48; (3) Training a Teacher network on professional F3-Set skeleton data; (4) Performing cross-domain unlabeled knowledge distillation to transfer structural motion priors to a Student RGB+Flow network; and (5) Few-shot adaptation on a small set of annotated campus videos.

Experimental results on our campus dataset, curated from public platforms, demonstrate that the proposed Skeleton-Guided Multimodal Distillation strategy significantly outperforms direct inference. By leveraging the invariant nature of skeletal features, the system effectively handles the "F3" (Fast, Frequent, Fine-grained) challenges in amateur videos, achieving a detection improvement comparable to state-of-the-art methods while maintaining high inference efficiency.

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Chapter 1 Introduction

1.1 Research Background and Significance

Tennis is a complex sport characterized by high tactical depth, rapid physical demands, and psychological intensity. In modern professional tennis, such as the ATP and WTA tours, computer vision-based analytics have become indispensable for performance enhancement. Technologies like Hawk-Eye and high-frame-rate tracking systems enable coaching teams to conduct fine-grained analyses of ball rotation, trajectory precision, and player biomechanics[1]. These insights drive tactical adjustments and injury prevention, representing the pinnacle of sports engineering.

However, a significant “deployment gap” exists between these elite environments and the vast landscape of collegiate and amateur tennis[2]. This gap is driven by four primary challenges:

Prohibitive Hardware Costs: Professional analytics rely on expensive multi-sensor arrays and high-performance GPU clusters, resources that are largely inaccessible to campus environments equipped only with standard monocular cameras or mobile devices.

Environmental Domain Shift: Unlike professional matches with perfect lighting and unobstructed views, campus courts are often plagued by fence obstructions, complex background clutter, and unstable lighting conditions.

Subject Variability: While professional players exhibit highly standardized movements, student players demonstrate high variability and non-standard actions, causing models trained on professional data to fail during transfer.

The F^3 Computational Challenge: Tennis events are defined as Fast, Frequent, and Fine-grained (F^3). Events occur in rapid succession (approximately one per second) with subtle visual distinctions[3], requiring precise timestamp localization that is difficult to achieve on edge devices.

With the advent of deep learning, particularly Knowledge Distillation (KD) and Multimodal Learning, bridging this gap is now possible. This research aims to democratize advanced sports analytics by developing a campus tennis action analysis system based on the MD-FED (Multimodal Distillation for Fine-grained Event Detection) framework. By leveraging the F^3 Set benchmark, this project seeks to transfer knowledge from professional broadcast data to amateur campus scenarios. This approach effectively breaks the data barrier between professional and amateur levels, providing a low-cost, high-precision intelligent solution for sports education.

1.2 Problem Statement

When applying existing advanced action recognition models to campus tennis scenarios, we face several core technical challenges:

The Perspective and Domain Discrepancy: The primary source of high-quality labeled data is the F^3Set , which consists of high-resolution broadcast footage from professional tournaments. These videos feature dynamic camera angles (zooms, pans, replays), variable lighting, and highly standardized professional player motions. In contrast, campus tennis footage is typically captured from a static, single-angle perspective (often behind the baseline), with cluttered backgrounds (fences, other courts) and non-standard amateur motions. Models trained purely on visual features (RGB) from broadcast data often fail to generalize to campus data, as they tend to overfit to specific background patterns rather than learning the essential motion dynamics.

Difficulty in Capturing F3 Features: Tennis exhibits typical F3 (Fast, Frequent, Fine-grained) characteristics. Actions are extremely fast (impact moments are often less than 0.1 seconds), frequencies are high (multiple shots per rally), and granularity is fine (e.g., distinguishing flat vs. topspin requires capturing minute wrist changes). Traditional Temporal Action Localization (TAL) models like ActionFormer often use sparse sampling or significant temporal downsampling, leading to the loss of critical instantaneous action details.

Data Scarcity and Annotation Bottleneck: Training high-performance models requires massive amounts of data with temporal labels. Although large-scale professional tennis datasets like F^3Set exist, acquiring and annotating fine-grained action data in campus scenarios is labor-intensive and costly. Existing end-to-end visual models struggle in these few-shot scenarios, where labeled data for the target domain is limited. Furthermore, due to the non-standard nature of amateur actions, models trained on professional data exhibit significant "Performance Degradation" on campus data. The system must be able to adapt to new environments (the campus court) without requiring massive re-annotation efforts.

Modality Absence and Compute Constraints: During the training phase, we can utilize multimodal information such as optical flow and skeletal pose to assist model learning. However, in the actual deployment (Inference) phase, limited by campus computing resources (such as standard laptops or mobile devices), inputs are often restricted to RGB video streams. Compressing "Privileged Knowledge" from multiple modalities into a single-modality model is the core difficulty of this study.

1.3 Research Objectives

This project aims to build and verify an end-to-end campus tennis action analysis system with the following specific objectives:

1. **Software and Pipeline Construction:** Based on Python and PyTorch frameworks, develop a complete computer vision pipeline. The system must integrate the F3ED architecture as the core detector and implement preprocessing modules for HRNet skeleton extraction and RAFT optical flow generation, capable of processing monocular RGB video streams and outputting fine-grained action event sequences.
2. **Exploration of Cross-Domain Distillation Methods:** Research and implement multimodal knowledge distillation strategies based on MD-FED. Construct a Teacher Ensemble containing RGB, Optical Flow (Motion), and Pose branches, and design heterogeneous loss functions (Feature Alignment) to transfer deep semantic features learned by the teacher model on professional data to the student model running on campus data.
3. **Quantitative Performance Assessment:** Establish a rigorous quantitative evaluation system. On F3Set (Source Domain) and the self-built Campus-NCAA (Target Domain) datasets, comprehensively evaluate the system’s detection accuracy and real-time performance using metrics such as Mean F1 Score, Edit Score, and Processing Frame Rate (FPS).
4. **Foundation for Future Development:** Build a modular, scalable system architecture to lay the code and theoretical foundation for introducing advanced functions such as trajectory tracking, 3D reconstruction, and federated learning in the future.

1.4 Scope and Contributions

This research primarily focuses on Stroke Detection and Fine-grained Classification in tennis, covering Serves, Forehands, Backhands, and their spin attributes (e.g., Topspin, Slice). The scope is limited to monocular camera perspectives (typically behind-the-baseline or rear-side views) and does not involve multi-camera synchronization or physical reconstruction of 3D ball trajectories.

The primary contributions of this work, corresponding to the stated objectives, include:

1. **Proposed MD-FED Cross-Domain Adaptation Scheme:** Applying the MD-FED multimodal distillation framework to the cross-domain scenario of sports action analysis, proving that using skeleton and optical flow as auxiliary modalities can significantly compensate for the generalization defects of RGB models on amateur data.
2. **Construction of High-Performance Teacher Ensemble:** Integrated HRNet-w48 (High-Resolution Pose Estimation) and RAFT (Recurrent All-Pairs Field Trans-

forms) as teacher branches. Compared to traditionally used OpenPose or TV-L1 optical flow, this significantly improved the teacher model’s ability to capture subtle actions and fast movements.

3. **Realization of an Extremely Efficient Student Model:** Established The optimized RGB single-modality student model achieved a success in reducing computation by approximately 75% compared to the multimodal teacher ensemble while maintaining over 90% of the detection precision, possessing high instructional utility.
4. **Validation of F3ED Architecture Effectiveness:** Validated the superiority of ”frame-level dense processing” and ”contextual refinement modules” in the F3ED architecture over traditional sparse sampling methods within the specific F3 scenario of tennis.

1.5 Report Organization

The remainder of this report is structured as follows:

Chapter 2 reviews the evolution of computer vision in sports analysis, focusing on F3 event detection, knowledge distillation, and skeletal pose estimation, clarifying the theoretical positioning of this study.

Chapter 3 details the technical implementation of the system, including teacher-student architecture design, loss function derivation, core detection algorithm logic, and presents qualitative and quantitative experimental results.

Chapter 4 deeply analyzes the technical mechanisms behind the experimental results, discusses method limitations, and analyzes the system’s application prospects in actual tennis teaching and future research directions.

Chapter 5 summarizes the entire work, reiterating research contributions and core value.

Chapter 2 Literature Review

This chapter presents a systematic review of computer vision methodologies applied to tennis action analysis. Starting with a survey of general advancements in the field, the discussion progressively focuses on sports-specific challenges, including temporal action localization and the detection of "F3" (Fast, Frequent, and Fine-grained) events. Key technical components, such as multimodal knowledge distillation, and skeletal pose estimation, are analyzed in depth. The chapter culminates by highlighting the scarcity of solutions for amateur and campus settings, establishing the primary research gap this study seeks to fill.

2.1 Recent Advancements in Computer Vision

The field of computer vision has undergone a revolution in the last decade, largely driven by the resurgence of deep learning. The success of deep convolutional neural networks (CNNs) on large-scale image recognition tasks, exemplified by AlexNet's victory in the 2012 ImageNet competition [4], marked a paradigm shift. This was followed by the development of even deeper and more sophisticated architectures like VGGNet [5], GoogLeNet [6], and ResNet [7], which introduced residual connections to enable the training of networks with hundreds of layers. These models have become the backbone of many modern computer vision systems.

More recently, the transformer architecture, originally developed for natural language processing [8], has been successfully adapted for computer vision tasks. The Vision Transformer (ViT) model demonstrated that a pure transformer-based architecture can achieve state-of-the-art results on image classification tasks, challenging the dominance of CNNs [9].

Beyond discriminative tasks, the last few years have seen extraordinary progress in generative models. Text-to-image models like DALL-E 2 [10], Imagen [11], and Stable Diffusion [12] can now generate highly realistic and complex images from natural language descriptions. Furthermore, the rise of large multi-modal models, such as GPT-4 [13], which can understand and process information from multiple modalities including text and images, is opening up new frontiers for computer vision and artificial intelligence in general. These advancements have significantly improved the performance of various computer vision tasks, including image classification, object detection, semantic segmentation, and image generation.

2.2 Applications of Computer Vision in Sports Analysis

The advancements in computer vision, particularly deep learning, have had a profound impact on sports analysis[1], driving the field from coarse-grained recognition to fine-

grained understanding[14].

Sports video analysis has evolved from Classification to Localization, and further to fine-grained Parsing. Early work focused on Action Classification, determining the sport type of a trimmed video clip. A seminal contribution in this era was the Temporal Segment Network (TSN) [15]. TSN introduced a sparse sampling strategy to model long-range temporal structures, enabling efficient learning from entire videos by aggregating information from distributed segments. This approach significantly improved classification accuracy by capturing both spatial appearances and temporal dynamics.

However, practical sports analytics, especially in tennis, require Temporal Action Localization (TAL)—precisely identifying the start and end times of actions in untrimmed long videos[16]. Architectures based on Transformers, such as ActionFormer [17], have become mainstream in the TAL field[18]. ActionFormer utilizes local self-attention mechanisms to model moments of actions without relying on pre-defined anchor windows, achieving state-of-the-art performance on general benchmarks like THUMOS14[17].

Yet, in the specific domain of tennis, these general-purpose TAL models face significant challenges. As defined by Liu et al. in the F3Set benchmark [19], tennis events are characterized as Fast, Frequent, and Fine-grained (F3). Critical moments, such as the "Ball Impact," often last less than 0.1 seconds (1-2 frames). General TAL models often employ excessive temporal downsampling to process long sequences[20], which leads to the loss of critical frame-level information. This results in large errors in locating instantaneous moments and difficulties in distinguishing subtle variations between rapid strokes (e.g., flat vs. slice serves) [19].

2.3 F^3 Event Detection: Fast, Frequent, and Fine-grained Challenges

The F^3 Set benchmark, proposed by Liu et al.[19], formally identifies the unique computational challenges in sports video analysis, defined as **Fast**, **Frequent**, and **Fine-grained** (F^3). Unlike generic video understanding tasks, tennis events present a rigorous test for temporal localization and attribute recognition:

- **Fast (Fast-paced Dynamics):** Tennis strokes are instantaneous events. The critical moment of ball-racket contact occurs within a micro-window, often spanning only 1-2 frames in standard broadcast video[21]. This results in significant motion blur and requires timestamp localization with high tolerance precision[22].
- **Frequent (High-Density Occurrence):** Events in a rally occur in rapid succession, averaging approximately one event per second (e.g., a serve followed immediately by a return and subsequent strokes). Models must distinguish between consecutive actions without losing context.

- **Fine-grained (Subtle Visual Distinctions):** The visual difference between semantically distinct events is minimal. For instance, distinguishing a "Slice" from a "Drop shot" or determining if a serve landed "Wide" versus "Body" requires analyzing subtle biomechanical cues and ball trajectories rather than just scene context[23].

2.3.1 Structured Definition of Tennis Events in $F^3\text{Set}$

To address these challenges, $F^3\text{Set}$ introduces a hierarchical, multi-attribute definition of tennis actions. An "event" is not merely a label (e.g., "hit") but a complex semantic combination. Specifically, a single tennis event is defined by the integration of **8 distinct sub-classes** ($sc_1, \dots, sc_8$), resulting in over 1,000 possible event types[24].

The structured definition includes:

1. **Player & Location** (sc_1, sc_2): Identifies who hit the ball (Near-end vs. Far-end player) and their court position (Deuce, Middle, or Ad court).
2. **Biomechanics** (sc_3, sc_6, sc_7): Captures the physical execution, including the side of the body (Forehand vs. Backhand), the specific shot technique (Groundstroke, Slice, Volley, Lob, Drop, Smash), and player movement (Approach the net vs. Non-approach).
3. **Tactical Attributes** (sc_4, sc_5): Defines the shot type (Serve, Return, Stroke) and the specific directional intent of the shot (T-area, Body, Wide, Cross-court, Down-the-line, Inside-out, etc.).
4. **Outcome** (sc_8): Classifies the result of the shot (In-bound, Winner, Forced Error, Unforced Error).

This granular definition allows for **Multi-level Granularity** analysis. Researchers can adjust the complexity from a coarse level (G_{low} , 38 event types) to a high-precision level (G_{high} , 1,108 event types), enabling the model to learn not just *when* an action occurred, but *how* and with *what result*.

2.3.2 The $F^3\text{ED}$ Solution

To handle this complexity, the $F^3\text{ED}$ model was proposed. It specifically addresses the "Deployment Gap" by abandoning traditional sparse sampling in favor of **Frame-wise Dense Prediction** combined with a **Multi-label Classification Head**. Unlike multi-class approaches that treat every combination as a unique class, $F^3\text{ED}$ predicts the probability of each sub-class element independently (e.g., predicting 'Forehand', 'Volley', and 'Cross-court' simultaneously), which effectively handles the combinatorial explosion of tennis events. This is further enhanced by a **Contextual Refinement Module**

(CTX) that learns logical sequences (e.g., a "Service" must be followed by a "Return"), correcting errors derived from pure visual processing[19].

2.4 Current Research on Multimodal Knowledge Distillation

In deep learning deployment, Knowledge Distillation (KD) has evolved from simple model compression to a sophisticated paradigm for cross-modal information transfer. In the context of sports analysis, where capturing rapid motion and fine-grained technique is critical[25], Cross-Modal Knowledge Distillation leverages the "Learning Using Privileged Information" (LUPI) strategy [26]. This paradigm allows a lightweight student model to hallucinate features from data modalities that are available during training but absent at inference time[27].

Recent advancements, such as the framework presented in Multimodal Distillation for Egocentric Action Recognition (ICCV 2023) [28], have formalized this approach by constructing a "Teacher Ensemble." This ensemble aggregates diverse expertise from distinct modalities—including RGB, Optical Flow, Audio, and Object Detections—and distills their joint posterior distributions (Logits) and intermediate feature maps to a unimodal "Student Model" (typically RGB-only).

This methodology is particularly valuable in sports scenarios for two reasons:

- **Temporal Dynamics and Motion Blur:** By distilling knowledge from an Optical Flow teacher, the RGB student learns to implicitly encode motion vectors. This is crucial for analyzing high-speed sports (e.g., tennis serves or swings) where motion blur often degrades pure RGB features [29].
- **Geometric Invariance:** In technique analysis, distilling features from a Skeleton/Pose-based teacher allows the student to focus on the geometric relationships of human joints, ignoring irrelevant background noise or clothing variations. For instance, Hong et al. [27] demonstrated that distilling pose information into RGB networks significantly improves fine-grained action recognition in sports like diving and tennis.

Consequently, the student model achieves "Privileged Information" capabilities, delivering the robustness of a multi-sensor system while maintaining the inference speed of a single-stream network[30]. This trade-off is essential for campus sports scenarios, enabling real-time, high-precision analytics on edge devices with limited computing power.

2.5 Skeleton-based Pose Estimation and HRNet Architecture

In action analysis, the human skeleton provides a high-level semantic representation robust to lighting, background, and clothing changes. HRNet (High-Resolution Network)

is a state-of-the-art model in the field of pose estimation[31]. Unlike traditional Hourglass or ResNet architectures, HRNet maintains high-resolution feature representations throughout the network depth. Through parallel multi-resolution subnetworks and repeated Multi-scale Fusion, it achieves pixel-level precise localization of human keypoints (e.g., wrists, ankles).

In tennis analysis, the 17 keypoints (COCO format) extracted by HRNet can construct a player’s Motion Skeleton Graph[32]. Based on this, Graph Convolutional Networks (like ST-GCN) can extract topological structural features of actions. However, the high computational complexity of HRNet (HRNet-w48 is approx. 32.9 GFLOPs) makes it difficult to run continuously in real-time on mobile devices[33], further validating the necessity of using it as an offline teacher model for distillation.

2.6 Research Gap and Proposed Approach

Despite significant strides in computer vision for sports analysis, particularly with the introduction of the F^3 ED model for handling fast, frequent, and fine-grained tennis events, critical limitations remain when transitioning from theoretical benchmarks to practical, amateur-level applications. This study identifies two primary research gaps and proposes a unified methodological solution.

2.6.1 The Domain Gap: From Professional Broadcasts to Campus Scenarios

Current state-of-the-art models, including F^3 ED, are predominantly trained and evaluated on high-quality datasets like F^3 Set, which feature professional broadcast footage with stable camera angles, high resolution, and standardized illumination. However, campus and amateur tennis scenarios present a distinct "wild" domain characterized by:

- **Visual Degradation:** Lower resolution, motion blur from handheld cameras, and variable lighting conditions.
- **Non-Standard Perspectives:** Unlike the fixed "center-court" broadcast view, campus videos often feature oblique angles or occlusions.
- **Performance Drop:** Preliminary research indicates that models trained on professional data suffer catastrophic performance degradation when applied to amateur footage (e.g., NCAA or campus datasets) due to this domain shift.

There is currently a lack of robust frameworks capable of transferring the fine-grained parsing capabilities of professional models to these resource-constrained, noisy campus environments.

2.6.2 Methodological Gap: Lack of Multimodal Distillation in Dense Tennis Prediction

While Multimodal Knowledge Distillation (KD) has proven effective in action recognition, existing frameworks like MD-FED have primarily been tailored for sparse classification tasks in egocentric views. There lacks existing adaptation of the MD-FED framework for third-person, dense temporal localization tasks in tennis. Specifically, the potential of using "privileged" teachers—such as Optical Flow for motion dynamics and HRNet-based Skeletons for biomechanical precision—to guide a lightweight F^3 ED student model remains unexplored.

2.6.3 Proposed Methodology: Cross-Domain Distillation for F^3 Event Detection

To address these gaps, this research proposes a novel integration of the MD-FED framework with the F^3 ED architecture. The core objective is to construct a domain-adaptive system that retains the rigorous event definition of professional analysis while being deployable in campus settings. The proposed approach involves:

1. **Adapting MD-FED for Third-Person Analysis:** We will modify the MD-FED "Teacher Ensemble" paradigm to suit third-person tennis video. This involves training expert "Teacher" networks on privileged modalities (Optical Flow and Skeleton heatmaps) that are robust to visual noise.
2. **Knowledge Distillation to F^3 ED Student:** These heavy teacher models will distill their knowledge (via logit-based and feature-based distillation) into a lightweight, RGB-only F^3 ED student model. This allows the student model to "hallucinate" motion and pose features during inference without the computational cost of extracting them.
3. **Campus Domain Generalization:** By leveraging the geometric invariance of the Skeleton teacher and the temporal robustness of the Optical Flow teacher, the RGB student model will learn features that are less sensitive to the specific visual style of professional broadcasts, thereby improving its generalization to amateur and campus video data.

This methodology aims to bridge the gap between high-end professional analytics and accessible campus sports technology, enabling real-time, fine-grained tennis analysis on standard consumer devices.

Chapter 3 Methodology, Experiments, and Results

This chapter details the technical implementation of the system, including system architecture, construction of teacher-student models, mathematical principles of distillation strategies, and core detection algorithm logic. It then presents experimental results based on F3Set and Campus-NCAA datasets.

3.1 System Overview and Architecture Design

The system adopts a modular pipeline design, with the overall architecture comprising five processing modules:

- **Data Input and Preprocessing Module:** This initial stage is responsible for interfacing with the video data. It handles the decoding of standard video formats (e.g., MP4, AVI), extracts individual frames, and performs essential preprocessing steps supporting professional broadcast video (25-30 FPS) and campus recorded video (60 FPS).
- **Frame Extraction and Pseudo-Label Generation:** The pipeline processes campus tennis videos in two distinct stages to bridge the domain gap. First, temporally segments the raw footage into individual rally clips and extracts frame-wise sequences, normalizing input resolutions. Then, deploys the F^3ED model (pre-trained on the professional F^3Set dataset) to perform inference on these extracted frames. This step automatically generates preliminary event predictions and timestamps (pseudo-labels) for the campus data, serving as a foundation for subsequent domain adaptation or knowledge distillation.
- **Multimodal Feature Extraction Module (Teacher Side - Stage 1):** An offline processing module used to generate high-quality training labels and features. Includes three parallel branches: RGB Visual Branch, RAFT Optical Flow Branch, and HRNet Skeleton Branch.
- **Cross-Domain Distillation and Training Module (Core Engine - Stage 2):** Utilizes Soft Targets and Feature Maps generated by the Teacher Ensemble to train a lightweight RGB Student Model via Feature Alignment Loss.
- **Real-time Detection and Inference Module (Student Side - Deployment):** Only the RGB Student Model is retained in the deployment phase. This model is based on the F^3ED architecture, performing frame-wise dense prediction and optimizing event sequences through a Bi-GRU contextual module to output final action types and timestamps.

3.2 Pipeline Design

The development of the system was guided by principles of modularity, extensibility, and computational efficiency. This section details the system architecture, the stages of the processing pipeline, and the rationale behind the implementation choices.

3.2.1 Program Architecture

The system is implemented in Python 3.8, utilizing the PyTorch deep learning framework to support complex model training and inference. The architecture adheres to a modular, multi-stage pipeline design pattern, specifically tailored for the complexities of domain adaptation in sports video analysis. By decoupling data preprocessing, feature extraction, and hierarchical model training, the system ensures reproducibility and scalability. The main components of the software are:

- **Rally Segmentation & Annotation Orchestrator (*process_manual_rallies.py* & *generate_annotations.py*):** This module serves as the foundational data ingestion layer, acting as the system’s entry point for raw video processing. It implements the logic for temporal segmentation, utilizing the pre-trained F^3ED model to intelligently identify and isolate valid tennis rally sequences from continuous raw footage, effectively filtering out non-play intervals. Following segmentation, it invokes the F^3ED model to generate preliminary, frame-level pseudo-annotations. This creates a structured metadata baseline (*rallies_metadata.json*) that maps raw video frames to their temporal contexts, which is critical for the subsequent supervised and semi-supervised training phases.
- **Biometric & Motion Feature Extraction Module (*batch_generate_skeleton.py* & *batch_generate_optical_flow.py*):** This module encapsulates the computationally intensive tasks of converting high-dimensional RGB data into semantic feature representations. It integrates two state-of-the-art vision backbones: HRNet (High-Resolution Net, w48) to extract high-fidelity 2D human skeletal keypoints, creating a skeleton-based representation of player biomechanics robust against background clutter; and RAFT (Recurrent All-Pairs Field Transforms) to compute dense optical flow fields[34], capturing temporal continuity and pixel-level motion patterns. By pre-computing these features, the system significantly reduces the overhead during the iterative training of action recognition models.
- **Source Domain Teacher Training Module (*run_md_fed_stage1.py*):** This module implements the initial supervised learning phase of the MD-FED framework. Its primary role is to establish a strong “Teacher” model using the labeled F^3Set tournament dataset (source domain). The module is configured to handle long-range temporal dependencies by utilizing an extended sequence length

(*clip_len=128*). It incorporates a customized training loop with warm-up epochs and gradient accumulation to ensure convergence on complex skeleton data, creating a robust baseline model that encapsulates the fundamental rules and motion patterns of professional tennis.

- Cross-Domain Knowledge Distillation Module (*train_md_fed_stage2.py*):**
 This component houses the core logic for Unsupervised Domain Adaptation (UDA), designed to bridge the gap between professional tournament footage and the target campus tennis environment. It employs a Teacher-Student architecture where the Stage-1 Teacher model is frozen. The module orchestrates a multi-modal distillation process, fusing the RGB-based student network with auxiliary Optical Flow information. By minimizing the discrepancy between the Teacher’s predictions and the Student’s outputs on the unlabeled campus dataset, the system forces the Student model to learn the semantic structure of tennis actions without requiring ground-truth labels for the new domain.
- Few-Shot Adaptation & Fine-Tuning Module (*few_shot_learning_stage3.py*):**
 This final processing module is responsible for the domain-specific refinement of the model. It loads the distilled Student model from Stage-2 and performs few-shot learning using a small subset of manually verified annotations from the campus dataset. This module not only fine-tunes the model weights to correct domain shift errors but also serves as an evaluation harness, facilitating a direct performance comparison between the initial zero-shot predictions and the fully adapted few-shot results to quantify the efficacy of the transfer learning pipeline.

This modular structure ensures a clean separation between generic feature extractors and domain-specific adaptation logic. It allows for the independent upgrading of backbone networks without disrupting the downstream distillation workflow, making the system highly extensible for future research into multi-modal action recognition.

3.2.2 Processing Pipeline Stages

The processing pipeline is designed as a sequential cascade of operations, transforming raw unstructured video data into actionable action recognition metrics through a multi-stage domain adaptation framework. The key stages are detailed below:

- Data Ingestion and Temporal Segmentation:** The pipeline initiates with the ingestion of raw, untrimmed broadcast footage from the campus tennis environment. Unlike standard datasets that offer pre-clipped actions, real-world data contains significant non-play intervals (e.g., player preparation, court changeovers). To address this, the system utilizes the pre-trained $F^3\text{Set}$ model configuration to

perform temporal localization. This stage iterates through the video stream, classifying frame sequences to distinguish active rallies from idle time, effectively generating a curated index of “Rally” clips. The output is a structured metadata file (*rallies_metadata.json*) that defines the temporal boundaries for all subsequent processing.

2. **Multi-Modal Feature Extraction:** To transcend the limitations of raw RGB pixel data and capture the intrinsic dynamics of tennis, this stage decomposes the video into two distinct semantic streams. First, the system employs **HRNet (w48)** to perform 2D pose estimation, extracting high-fidelity skeletal keypoints for each player. This abstracts away background noise and focuses on biomechanical movements. Simultaneously, the **RAFT** model is invoked to compute dense optical flow maps, capturing the pixel-level motion vectors and temporal continuity between consecutive frames. These extracted features—Skeleton and Optical Flow—are serialized and stored, serving as the fundamental inputs for the multi-modal distillation process.
3. **Source Domain Teacher Initialization (Stage-1):** Before tackling the target domain (campus tennis), the pipeline constructs a robust knowledge prior. This stage involves training a “Teacher” model solely on the fully labeled F3-Set professional tournament dataset. The training logic is optimized for long-sequence action recognition (using a specific clip length and stride), allowing the model to learn complex temporal dependencies inherent in professional play. This supervised learning phase establishes a high-performance baseline, encapsulating the generalized rules and stroke patterns of tennis, which will serve as the reference standard for the subsequent adaptation stages.
4. **Cross-Domain Knowledge Distillation (Stage-2):** This is the most computationally sophisticated stage of the pipeline, where Unsupervised Domain Adaptation (UDA) takes place. The pre-trained Teacher model is frozen to preserve its learned representations. The system then instantiates a “Student” model and trains it on the unlabeled campus tennis data (Target Domain). By leveraging the multi-modal inputs prepared in Stage 2, the pipeline executes a distillation protocol where the Student model minimizes the divergence between its predictions and the Teacher’s soft labels. This effectively forces the Student to align its feature space with the Teacher’s, learning to recognize actions in the campus environment without requiring explicit ground-truth labels.
5. **Few-Shot Refinement and Inference (Stage-3):** In the final stage, the system bridges the residual domain gap through supervised fine-tuning. A small, manually verified subset of the campus dataset (Few-Shot samples) is injected into the

training loop. The distilled Student model undergoes a final round of optimization to correct specific domain shifts that unsupervised methods may have missed. The output of this stage is the fully adapted inference model, which is then used to predict action classes on the test set. Performance metrics are calculated by comparing these predictions against the manual annotations, quantifying the system’s accuracy and robustness in the target environment.

3.2.3 Implementation Choices and Justifications

The selection of the technological stack and algorithmic components was driven by the imperative to balance computational efficiency with the high precision required for fine-grained action recognition. The system is architected primarily in Python, leveraging its dominant position in the deep learning ecosystem to integrate disparate state-of-the-art libraries. The detailed justifications for the core components are as follows:

- **PyTorch Framework:** PyTorch was selected as the foundational deep learning framework due to its dynamic computation graph (define-by-run) paradigm. This feature is crucial for the complex, multi-stage training pipeline of MD-FED, allowing for flexible debugging and runtime modifications of the Teacher-Student distillation architecture. Its extensive model zoo and active research community also facilitated the seamless integration of pre-trained backbones like HRNet and RAFT, significantly accelerating the development cycle compared to static graph frameworks.
- **High-Resolution Network (HRNet):** For skeletal feature extraction, we employed HRNet (specifically the w48 variant) rather than traditional ResNet-based pose estimators. The justification lies in HRNet’s ability to maintain high-resolution representations throughout the network depth, rather than recovering resolution from low-resolution feature maps. In the context of tennis analysis, where limb positioning and subtle body mechanics are critical for distinguishing stroke types (e.g., forehand vs. backhand), this preservation of spatial granularity is essential for accurate keypoint localization.
- **RAFT (Recurrent All-Pairs Field Transforms):** To capture temporal dynamics, we utilized RAFT for optical flow estimation. Unlike classical methods (e.g., Farneback) or earlier deep learning approaches (e.g., FlowNet), RAFT employs a recurrent GRU-based update operator that iteratively refines the flow field. This choice is justified by the fast-paced nature of tennis; RAFT demonstrates superior robustness in handling large displacements and motion blur—common artifacts in amateur broadcast footage—ensuring that the motion stream provides reliable cues for the action recognition model.

- **Multi-Modal Knowledge Distillation (MD-FED):** The decision to implement a multi-modal distillation framework was driven by the scarcity of labeled data in the target domain (campus tennis). Developing a fully supervised model from scratch would require prohibitive manual annotation efforts. By adopting an Unsupervised Domain Adaptation (UDA) strategy, specifically the MD-FED approach, we effectively leverage the rich, existing annotations from the professional F^3 Set dataset. This methodology allows the system to transfer learned semantic knowledge to the campus domain without direct supervision, offering a scalable solution to the domain shift problem.
- **OpenCV & NumPy Integration:** While deep learning handles the semantic analysis, OpenCV serves as the robust backend for video I/O and basic image preprocessing (resizing, normalization). NumPy is employed for all tensor-agnostic numerical operations, providing efficient, vectorized array manipulations that bridge the gap between raw video frames and the tensor formats required by the PyTorch models.

3.3 The Core Detection Algorithm: Frame-by-Frame Event Analysis

At the heart of the system lies the F^3ED (Fast Frequent Fine-grained Event Detection) algorithm, which processes each segmented video clip to identify and classify complex tennis event sequences [19]. Unlike traditional action recognition methods that assign a single label to a coarse video segment, this algorithm is a carefully orchestrated sequence of deep learning modules designed to handle the rapid temporal dynamics and subtle visual distinctions inherent in sports analytics. The pipeline is robust to motion blur and occlusion, leveraging end-to-end training to optimize detection accuracy. This section provides a detailed, step-by-step breakdown of the algorithmic components.

3.3.1 Spatiotemporal Feature Extraction

The first step in the analysis of any given video segment is to transform raw pixel data into a high-level semantic representation capable of capturing both spatial details (e.g., court position) and temporal motion (e.g., swing mechanics).

Video Encoder Construction. The system employs a frame-wise feature extraction strategy rather than processing non-overlapping clips, ensuring high temporal precision [35]. The backbone architecture utilizes a 2D CNN, specifically the **RegNet-Y 200MF** [36], augmented with a **Temporal Shift Module (TSM)** [37].

The TSM is critical for efficiency; it facilitates the exchange of information between neighboring frames by shifting a portion of the channels along the temporal dimension. This allows the model to capture spatiotemporal features with the computational efficiency of 2D convolutions, avoiding the high cost of 3D convolutions [37]. Formally, let $X \in \mathbb{R}^{H \times W \times 3 \times N}$ denote the input video clip consisting of N RGB frames. The Video Encoder (VE) processes these frames to produce a sequence of feature embeddings F_{emb} :

$$F_{emb} = VE(X), \quad \text{where } F_{emb} \in \mathbb{R}^{N \times d'} \quad (1)$$

Here, d' represents the dimension of the feature vector. To further capture long-term visual dependencies beyond the local receptive field of the CNN, the encoder output is processed by a bidirectional Gated Recurrent Unit (BiGRU) [38], reducing the high-dimensional video data into a compact sequence of feature vectors ready for specific detection tasks.

3.3.2 Dense Event Localization (LCL)

A critical challenge in analyzing fast-paced sports is distinguishing the exact moment of an event (e.g., ball impact) from background frames. The system employs an **Event Localizer (LCL)** module to perform dense binary classification on the extracted features.

The LCL consists of a fully connected network culminating in a Sigmoid activation function. It iterates through the feature sequence F_{emb} , assigning a probability score to each frame indicating the likelihood of an event occurrence. For an N -frame clip, the predicted probability sequence $(\hat{p}_1, \dots, \hat{p}_N)$ is computed as:

$$(\hat{p}_1, \dots, \hat{p}_N) = \text{Sigmoid}(LCL(F_{emb})) \quad (2)$$

During training, the discrepancy between predicted probabilities and ground truth binary labels $p_i \in \{0, 1\}$ is minimized using the binary cross-entropy loss function:

$$L_{LCL} = \frac{1}{N} \sum_{i=1}^N [p_i \cdot \log(\hat{p}_i) + (1 - p_i) \cdot \log(1 - \hat{p}_i)] \quad (3)$$

To address the extreme class imbalance—where event frames constitute less than 3% of the total footage—the loss weight for foreground classes ($p_i = 1$) is significantly increased (e.g., fivefold) relative to background classes [23].

3.3.3 Multi-Label Event Classification (MLC)

Once potential event timestamps are identified, the system must categorize the specific attributes of the event. Given the complexity of tennis actions, which involve combina-

torics of shot type, direction, and outcome (resulting in over 1,000 possible event types), a standard multi-class approach is inefficient and suffers from long-tail distribution issues.

Attribute Vectorization. Instead, the algorithm utilizes a **Multi-Label Classifier (MLC)**. Each event E_i is represented as a binary vector $[e_{i,1}, \dots, e_{i,K}]$ of length K , where each element corresponds to the presence of a specific attribute (e.g., “Forehand”, “Cross-court”, “Winner”). The MLC maps the feature vector f_i at the detected timestamp to this attribute space:

$$\hat{E}_i = \text{Sigmoid}(\text{MLC}(f_i)) = [\hat{e}_{i,1}, \dots, \hat{e}_{i,K}] \quad (4)$$

This formulation mitigates the combinatorial explosion problem. The optimization minimizes the multi-label classification loss L_{MLC} averaged over M detected events:

$$L_{MLC} = \frac{1}{M} \sum_{i=1}^M \left(\frac{1}{K} \sum_{j=1}^K [e_{i,j} \cdot \log(\hat{e}_{i,j}) + (1 - e_{i,j}) \cdot \log(1 - \hat{e}_{i,j})] \right) \quad (5)$$

3.3.4 Contextual Sequence Refinement (CTX)

The final stage addresses the limitations of purely visual processing. Rapid motion blur or occlusion can lead to individual frame misclassifications that are logically inconsistent (e.g., a “Service” event appearing in the middle of a rally, or a player hitting a “Forehand” from a “Backhand” stance).

To rectify this, the system incorporates a **Contextual Module (CTX)**. This module treats the sequence of predicted event vectors $\hat{E}$ as a time-series and employs a bidirectional GRU [39] to model the causal relationships and transition probabilities between consecutive events. The refinement process is defined as:

$$(\mathbb{E}_1, \dots, \mathbb{E}_M) = \text{CTX}(\hat{E}_1, \dots, \hat{E}_M) \quad (6)$$

The output is a refined sequence of event representations $\mathbb{E}_i$ that integrates both the initial visual predictions and the learned contextual logic. The module is trained using a similar cross-entropy loss L_{CTX} on the refined outputs. This end-to-end refinement significantly improves the coherence of generated annotations, ensuring the final output adheres to the logical flow of the sport [19].

3.4 Multimodal Distillation Framework (MD-FED)

The core of the proposed system is the Multimodal Distillation for Few-Shot Fine-Grained Event Detection (MD-FED) framework. Addressing the challenge of adapting models to the college tennis domain where labeled data is scarce, this framework employs a weakly

supervised, three-stage training strategy. It leverages the structural invariance of human skeletons (Teacher) to guide the learning of rich visual representations (Students) from RGB and Optical Flow modalities. This approach allows the system to generalize effectively to new environments by distilling motion knowledge into visual encoders without requiring massive annotations.

3.4.1 Stage I: Domain-Specific Skeleton Feature Pretraining

The first stage involves establishing a robust “Teacher” model capable of capturing the fundamental biomechanics of tennis strokes. Unlike pixel-based features which are susceptible to background noise and lighting variations, skeletal representations provide a compact and distinct signal of human motion.

- **Input Representation:** The system utilizes off-the-shelf estimators (e.g., HRNet [31]) to extract 2D human pose sequences $X_{pose} \in \mathbb{R}^{T \times P \times V \times 2}$ from the source domain videos, where T is the temporal length, P is the number of persons, and V is the number of joints.
- **Teacher Architecture:** The skeleton feature extractor ($\mathcal{E}^{teacher}$) is built upon a spatial-temporal graph convolutional network, specifically **STGCN++** [40], followed by a bidirectional Gated Recurrent Unit (BiGRU) [38]. This architecture is designed to model both the spatial configuration of limbs and the long-term temporal dependencies of a rally.
- **Optimization:** The teacher is trained on the labeled source dataset using a composite loss function comprising the Event Localization Loss ($\mathcal{L}_{LCL}$) and Event Classification Loss ($\mathcal{L}_{CLF}$):

$$\mathcal{L}_{stage1} = \mathcal{L}_{LCL} + \mathcal{L}_{CLF} \quad (7)$$

Here, $\mathcal{L}_{LCL}$ utilizes binary cross-entropy to distinguish event frames from background, while $\mathcal{L}_{CLF}$ ensures accurate categorization of stroke types (e.g., Forehand vs. Backhand).

3.4.2 Stage II: Cross-Modal Knowledge Distillation

This stage constitutes the core contribution of the MD-FED framework. It aims to transfer the domain-invariant motion knowledge learned by the Teacher into the visual modalities (RGB and Optical Flow), which contain richer contextual information (e.g., ball trajectory, racket contact) but are harder to train on small datasets.

- **Student Architectures:** We initialize two “Student” networks: an RGB encoder ($\mathcal{E}_{RGB}^{student}$) and an Optical Flow encoder ($\mathcal{E}_{flow}^{student}$). These networks employ a **TSM**

(**Temporal Shift Module**) [37] integrated with a **RegNet-Y 200MF** backbone [36]. This choice ensures efficient spatiotemporal feature extraction with a controlled parameter budget.

- **Distillation Mechanism:** During this phase, the Teacher network is frozen. The Student networks are trained on a large set of unlabeled data from the target domain. The objective is to force the students’ feature embeddings F_m to align with the teacher’s skeletal embeddings F_{pose} in a shared high-dimensional latent space.
- **Loss Function:** The training minimizes the L_2 distance (Mean Squared Error) between the teacher’s and students’ feature vectors:

$$\mathcal{L}_{d_m} = \|F_{pose} - F_m\|_2^2, \quad m \in \{RGB, flow\} \quad (8)$$

By minimizing this distillation loss, the visual encoders learn to ignore background clutter and focus on the motion dynamics that correlate with the skeleton, effectively performing self-supervised domain adaptation.

3.4.3 Stage III: Few-Shot Target Domain Adaptation

The final stage refines the distilled Student networks to perform precise event detection in the specific context of the college tennis environment, using only a limited number of labeled samples (Few-Shot).

- **Fine-Tuning:** The pre-trained RGB and Optical Flow encoders from Stage II are connected to the downstream Event Detector heads (Localizer and Classifier).
- **Optimization:** The entire student pipeline is fine-tuned on the k -clip few-shot dataset (D_{label}) from the target domain. The loss function mirrors Stage I, optimizing for detection accuracy:

$$\mathcal{L}_{stage3} = \mathcal{L}_{LCL} + \mathcal{L}_{CLF} \quad (9)$$

This stage allows the model to calibrate its predictions based on the specific camera angles and visual characteristics of the college courts, combining the general motion robustness acquired from the Teacher with the specific visual details learned from the target data.

3.5 Experimental Setup and Quantitative Assessment

This section details the experimental methodology used to evaluate tennis event detection system. It describes the dataset, the process for assessing detection accuracy in the

absence of a formal ground truth, the metrics used for quantitative evaluation, and the computational environment in which the experiments were conducted.

3.5.1 Dataset

The evaluation of our approach is conducted on a comprehensive dataset that integrates professional match footage from the F^3Set benchmark with newly collected college-level tennis videos. This combination ensures that the proposed model is evaluated under both elite professional conditions and diverse real-world collegiate scenarios, thereby improving robustness and generalizability.

Professional Data (F^3Set) The F^3Set (Tennis Single) benchmark consists of 114 high-resolution broadcast matches featuring 75 professional players. It contains 11,584 rallies and a total of 42,846 precisely timestamped shot events. The dataset provides fine-grained annotations across 1,108 event types organized into 8 sub-classes, covering diverse court surfaces, match environments, and lighting conditions.

College-Level Data (Campus Tennis) To further extend the diversity of the benchmark, we collected and annotated 7 additional college-level tennis videos. This campus dataset contains 96 annotated rally clips with a total duration of 915.37 seconds, corresponding to 54,945 frames and 460 fine-grained events. All videos were recorded at high frame rates (predominantly 60 FPS) to preserve temporal precision for fast ball–racket interactions. A detailed statistical summary of each campus video sequence is provided in Table 1.

Table 1: Statistical Summary of the Campus Tennis Dataset

Video ID	Rallies	Total Frames	FPS	Duration (s)	Events
6VSmpCSgY7M	14	5,928	60.02	98.76	32
Avendano vs Penzlin	13	10,251	60.02	170.78	95
dwPey52i1LE	11	8,511	60.03	141.79	80
Hoole vs Dong	12	5,751	60.02	95.81	47
IohTeru65U4	13	5,270	60.03	87.80	40
Lc9MSf6vHxU	17	7,305	60.02	121.71	61
RUokidaZR30	16	11,929	60.03	198.73	105
Total	96	54,945	–	915.37	460

3.5.2 Annotation and Event Distribution

All campus tennis videos were annotated following a rigorous three-stage verification process. Each rally was manually reviewed, and events were annotated at the frame level to accurately identify ball–racket contact moments. Across the complete campus

dataset, we annotated 96 video segments containing 460 events and 54,945 frames, with an average of 4.79 events per video. Video lengths range from 179 to 1,800 frames, reflecting significant variability in rally dynamics.

The annotation schema follows the F^3Set standard and covers eight fine-grained sub-classes. Statistical analysis shows that the campus dataset exhibits a well-balanced distribution across multiple dimensions:

Table 2: Statistical Distribution of Manual Annotations

Sub-Class	Element	Count	Proportion (%)
SC1	near	228	49.6
	far	232	50.4
SC2	deuce	144	31.3
	ad	157	34.1
	middle	159	34.6
SC3	forehand	204	56.5
	backhand	157	43.5
SC4	serve	96	21.0
	return	65	14.2
	stroke	296	64.8
SC5	T	35	7.7
	Wide	58	12.7
	cross-court	160	35.2
	down the line	44	9.7
	down the middle	91	20.0
	inside-in	37	8.1
	inside-out	30	6.6
SC6	ground stroke	304	83.5
	slice	24	6.6
	volley	12	3.3
	lob	9	2.5
	drop	13	3.6
	smash	2	0.5
SC7	approach	29	6.3
	non-approach	431	93.7
SC8	in-bound	363	79.1
	winner	22	4.8
	forced error	48	10.5
	unforced error	26	5.7

Ethical Standards Both F^3Set and the campus tennis dataset are constructed exclusively from publicly available sports broadcasts and recordings. We do not redistribute video content and only provide reference links to ensure compliance with copyright regu-

lations. All data focuses solely on on-court athletic performance and is used strictly for academic research purposes.

3.5.3 Ground Truth and Validation Approach

A critical requirement for evaluating the proposed system is the availability of high-quality ground truth annotations. Unlike previous iterations of this project that relied on subjective qualitative assessment, this work adopts a rigorous, frame-level ground truth dataset composed of both professional benchmark data and custom-annotated collegiate match footage.

Professional Dataset (F^3Set) For the professional component of the evaluation, we leverage the existing ground truth annotations provided by the F^3Set benchmark. These annotations were produced by a team of domain experts through a multi-stage pipeline that combines automated clip selection with manual frame-by-frame verification. Each event in F^3Set is defined as an instantaneous moment of ball–racket contact, with timestamps recorded at frame-level precision and a tolerance of 1–2 frames. The ground truth further includes detailed labels across eight sub-classes, covering attributes such as shot direction, technique, and outcome.

Campus Tennis Dataset (Manual Annotation) As no pre-existing annotations are available for the campus-specific data, all collegiate match footage was manually annotated by the author following the strict guidelines and hierarchical label taxonomy established by F^3Set . Using an interactive annotation toolchain, each rally was reviewed on a frame-by-frame basis to identify the exact moment of ball contact. For each detected event, labels were assigned across multiple dimensions, including:

- **Temporal Localization:** Identifying the precise frame corresponding to ball–racket contact.
- **Multi-level Granularity:** Assigning fine-grained elements such as player position, court location, and shot technique (e.g., ground stroke, slice, or volley).
- **Result Verification:** Labeling event outcomes, including in-bound shots, winners, and various error types, to ensure consistency with professional annotation standards.

Validation Strategy The system’s performance is quantitatively evaluated by comparing model predictions against the ground truth timestamps and semantic labels. This shift from subjective visual inspection to an objective, benchmark-aligned validation protocol enables a more rigorous assessment of detection accuracy and labeling reliability,

while also measuring the model’s ability to generalize across different skill levels and recording environments.

3.5.4 Evaluation Metrics

To quantitatively characterize the system’s performance in detecting fast, frequent, and fine-grained tennis events, we employ the following evaluation metrics:

Mean F1-Score with Temporal Tolerance This metric serves as the primary criterion for assessing both classification accuracy and temporal localization precision. A prediction is counted as a True Positive (TP) only if its predicted timestamp falls within a strict temporal tolerance window (e.g., ± 1 frame) of the ground truth event and the associated class label is correctly identified. This formulation explicitly enforces temporal alignment in addition to semantic correctness.

$F1_{evt}$ and $F1_{elm}$ Given the long-tail distribution of the 1,108 fine-grained event types, two complementary variants of the F1-score are reported to ensure a balanced evaluation. $F1_{evt}$ denotes the mean F1-score averaged across all distinct event types, reflecting the model’s performance on rare and frequent events alike. In contrast, $F1_{elm}$ represents the mean F1-score averaged across all individual semantic elements (e.g., *forehand*, *backhand*, *in-bound*), which typically follow a more balanced distribution and provide insight into element-level recognition robustness.

Edit Score The Edit Score is computed based on the Levenshtein distance between the predicted and ground truth event sequences. This metric measures the structural similarity of event orderings and penalizes missing, extra, or misordered events. It is particularly important for evaluating the model’s understanding of the temporal logic and coherence of an entire rally.

Processing Efficiency To assess computational efficiency, the per-frame inference time of the proposed F^3ED model is measured in milliseconds (ms). This metric is crucial for determining the feasibility of deploying the system in real-time or near-real-time sports analytics applications.

3.5.5 Experimental Environment

All experiments were conducted in a high-performance deep learning environment optimized for large-scale video-based temporal action understanding. The detailed hardware and software configurations are described below.

Hardware Specifications All models were trained and evaluated on a workstation equipped with an NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM. The system supports a peak power capacity of 450 W, ensuring stable performance during intensive end-to-end fine-tuning of video encoders.

Software and Driver Configuration The experimental platform runs on a Linux-based operating system with NVIDIA Driver version 550.163.01 and CUDA Toolkit 12.4. The project was implemented using the following core software libraries:

- **Deep Learning Framework:** PyTorch 2.9.1 with TorchVision 0.24.1.
- **Video Processing:** OpenCV 4.12.0 and PySceneDetect 0.6.7 for automated shot transition detection.
- **Optimization:** AdamW optimizer implemented in PyTorch, with DeepSpeed 0.18.4 for memory-efficient training.
- **Scientific Computing:** NumPy 2.2.6, SciPy 1.16.3, and Scikit-learn 1.8.0 for data preprocessing and metric computation.

Training Protocol All models were trained using a cosine annealing learning rate schedule, with an initial learning rate of 10^{-3} for lightweight encoders (e.g., TSM) and 10^{-4} for more complex architectures (e.g., SlowFast). Training was performed for 500 epochs, with each epoch taking approximately 1 minute on the dedicated GPU.

3.6 Results and Analysis

This section presents the experimental results of the proposed model evaluated on the integrated dataset. The system performance is analyzed from both quantitative and qualitative perspectives. Quantitative evaluation is conducted using the standard metrics described in Section 3.5.4, while qualitative analysis examines the consistency and realism of predicted event sequences relative to the ground truth.

3.6.1 Quantitative Results

The proposed system is evaluated against baseline configurations, with particular emphasis on the performance gains achieved in the final stage of model development (Stage 3). A comparative summary between the initial baseline model result and our result is reported in Table 3.

The experimental results demonstrate that the Stage 3 model consistently outperforms the baseline across all evaluation metrics. In particular, the system achieves a substantial improvement in *element-level recognition* ($F1_{elm}$), increasing from 1.9% to

Table 3: Comparative performance results on the Tennis Single dataset.

Metric	F^3ED	Our	Improvement
$F1_{evt}$	0.366	0.433	+18.3%
$F1_{elm}$	0.019	0.110	+478.9%
Mean F1 (± 1 frame)	0.097	0.180	+85.6%
Edit Score	13.29	40.23	+202.8%

11.0%. This nearly **5.8-fold improvement** highlights the effectiveness of the proposed architecture in modeling fine-grained semantic elements under a long-tail distribution.

Event Localization and Element Classification The Stage 3 model achieves an $F1_{evt}$ score of 43.3%, representing an 18.3% relative improvement over the baseline. This gain indicates improved sensitivity in detecting precise event timestamps under strict temporal tolerance constraints. Furthermore, the Mean F1 score improves by 85.6%, reaching 18.0%, reflecting superior performance in recognizing individual action elements such as shot technique and player position.

Sequence Similarity The Edit Score increases significantly from 13.29 to 40.23 (+202.8%). This improvement confirms that the predicted event sequences generated by the Stage 3 model are substantially more aligned with the temporal structure and causal logic of real-world tennis rallies compared to earlier iterations.

3.6.2 Qualitative Analysis and Main Findings

The qualitative performance gains observed in Stage 3 are primarily attributed to the integration of the multi-label classification head and the contextual modeling module. Several key improvements are consistently observed:

- **Logical Consistency:** Unlike baseline models that occasionally produce invalid event sequences (e.g., a *winner* followed by another shot), the Stage 3 model leverages contextual correlations to enforce realistic event-wise causal relationships.
- **Fine-grained Recognition:** The model exhibits enhanced discrimination between visually similar actions, such as distinguishing a *slice* from a *drop shot* in fast-paced collegiate video clips.
- **Structural Accuracy:** The predicted sequences more accurately reflect the natural rhythm of tennis rallies, successfully capturing fast and frequent events occurring at a rate of approximately one event per second.

3.6.3 Performance and Real-Time Capability

Computational efficiency remains a key strength of the proposed system. When deployed on a single NVIDIA GeForce RTX 4090 GPU, the model achieves a fast average per-frame inference time. This processing speed supports real-time or near-real-time deployment in practical scenarios such as sports broadcasting and tactical analysis, enabling timely feedback for coaches and players even when processing high-resolution video streams.

3.7 Chapter Summary

This chapter detailed the core technical path of the College-Tennis system. By constructing a high-performance Teacher Ensemble with RAFT and HRNet and utilizing cross-domain distillation, we successfully compressed multimodal knowledge into a lightweight F3ED Student Model. Results prove this method achieves great real-time performance and 44.3% F1 accuracy on amateur video, effectively overcoming domain shift.

Chapter 4 Discussion and Future Directions

This chapter provides an in-depth discussion of the proposed system, synthesizing experimental findings with methodological insights.

4.1 Key Findings and Research Value

A central conclusion of this study is that high-fidelity, fine-grained action understanding does not inherently require expensive inference-time computation. Instead, through carefully designed knowledge distillation, professional-level temporal priors can be transferred into lightweight models suitable for educational environments.

4.1.1 Bridging Professional Domains to Campus Scenarios

The practical significance of this work lies in demonstrating that models trained on elite professional tournaments (e.g., F^3Set) can generalize effectively to resource-constrained campus environments.

- **Transfer of Professional Priors:** The F^3Set benchmark encodes rich athletic knowledge derived from top-tier players, including precise temporal patterns, biomechanical regularities, and tactical structures. Through distillation, these priors are absorbed by the student model without explicit access to multi-modal inputs.
- **Robustness under Domain Shift:** Despite substantial differences between professional broadcasts and campus recordings—such as background clutter, non-standard techniques, and heterogeneous camera viewpoints—our system maintains stable performance, indicating strong cross-domain robustness.
- **Empirical Validation:** On the campus dataset, the Stage 3 model achieves a **202.8% improvement in Edit Score** over the baseline, confirming that professional temporal logic can be successfully transferred to amateur scenarios.

4.1.2 Balancing High Performance and Lightweight Deployment

Traditional fine-grained video recognition systems often depend on heavy multi-modal pipelines and high computational budgets, limiting their applicability in educational settings.

- **Elimination of Explicit Modalities:** Our system removes the need for optical flow or pose estimation at inference time, significantly reducing computational overhead.

- **Implicit Perception:** By leveraging feature alignment during training, the student model learns motion-sensitive representations implicitly, achieving an ultra-low inference latency on a single RTX 4090 GPU.

This balance enables real-time deployment without sacrificing recognition fidelity.

4.2 Technical Insights and Mechanistic Analysis

4.2.1 Implicit Perception and Feature Mimicry

One of the most important technical insights of this work is the effectiveness of *Implicit Perception*. Although the student model operates solely on RGB inputs during inference, feature alignment losses force its convolutional kernels to evolve toward motion-aware representations.

Guided by a high-capacity teacher, the student network learns to attend to human joints, limb boundaries, and interaction regions, effectively mimicking the perceptual sensitivity of pose- and flow-based models. This demonstrates that deep neural networks possess strong feature mimicry capabilities and can exhibit perceptual depth exceeding the raw information available in RGB frames alone.

4.2.2 Contextual Reasoning and Logical Error Correction

In unconstrained campus environments, visual ambiguity is common due to non-standard techniques and occlusions. The Contextual Module (CTX) plays a crucial corrective role by introducing tennis-specific temporal logic into the prediction process.

- **Structural Prior:** CTX encodes causal constraints such as the canonical sequence of serve, return, and stroke, acting as a Bayesian prior over event sequences.
- **Error Tolerance:** By enforcing logical consistency, CTX corrects visually plausible but semantically invalid predictions (e.g., a winner followed by another shot).

This reasoning mechanism significantly improves top-1 accuracy and sequence coherence in real-match scenarios.

4.3 Strengths and Limitations

4.3.1 Strengths

Interpretability via Proxy Teacher Branches Although the student model remains a high-speed black box, the teacher’s multi-branch structure (RGB, Pose, and Flow) provides transparency during training and debugging. Errors can be traced back to specific perceptual failures, facilitating targeted refinement.

Data Efficiency and Transferability Our system substantially reduces annotation requirements by distilling knowledge from professional datasets. This makes rapid adaptation to new campuses or even new sports feasible with minimal labeled data.

Low-Latency Edge Deployment The distilled model achieves real-time inference on commodity hardware, democratizing access to advanced sports analytics for schools with limited computational resources.

4.3.2 Limitations

Teacher Dependency and Negative Transfer The student’s performance ceiling is inherently bounded by teacher quality. If teacher models fail under extreme lighting or noisy conditions, incorrect priors may be transferred, leading to persistent errors.

Viewpoint Sensitivity The current system is sensitive to extreme camera perspectives uncommon in professional broadcasts. Viewpoint-invariant representation learning remains an open challenge.

4.4 Future Research Directions

4.4.1 Test-Time Adaptation

Dynamic adaptation of normalization statistics will be explored to improve robustness across varying lighting, court colors, and weather conditions without retraining.

4.4.2 Multi-modal 3D Reconstruction and Ball Tracking

Integrating a ball-tracking branch and 3D trajectory reconstruction could enable finer discrimination between visually similar but physically distinct actions.

4.4.3 Privacy-Preserving Federated Learning

A federated framework would allow institutions to collaboratively improve models by sharing gradients rather than raw video data, ensuring privacy while enhancing generalization.

4.5 Chapter Summary

This chapter analyzed the system from both methodological and practical perspectives. By shifting from explicit to implicit perception through distillation, the proposed approach achieves a rare balance between computational efficiency and fine-grained recognition accuracy. While challenges such as teacher dependency and viewpoint sensitivity

remain, MD-FED provides a scalable pathway for transferring elite sports intelligence into real-world educational settings.

Chapter 5 Conclusion

This work presents the **Detection of Fine-Grained Tennis Events in Campus Videos via Skeleton-Guided Multimodal Distillation** project, which addresses the challenge of enabling fine-grained tennis action analysis in resource-constrained campus environments. By integrating the proposed **MD-FED multimodal distillation framework** with the **F³ED fine-grained event detection architecture**, this study demonstrates that professional-level athletic priors can be effectively transferred to amateur scenarios using a lightweight, single-modality inference model.

Quantitative experiments on campus tennis videos show consistent and meaningful improvements over a directly transferred professional baseline. In particular, the proposed Stage 3 model improves the **Edit Score** from 13.29 to 40.23, representing a **202.8% relative gain**. This result indicates a qualitative transition from fragmented, logically inconsistent predictions to temporally coherent event sequences that better reflect the intrinsic structure of real tennis rallies. Such structural consistency is critical for downstream analysis tasks, including match summarization and tactical assessment.

At finer temporal resolutions, the model achieves a more than **five-fold improvement** in event-level F1 score (from 0.019 to 0.110) and an **85.6% increase** in element-level F1 score (from 0.097 to 0.180). Although the absolute values of these metrics remain modest, this outcome is expected given the extreme difficulty of instantaneous hit detection and fine-grained action recognition in amateur tennis videos, which often suffer from motion blur, occlusion, non-standard techniques, and limited ball visibility. Within this challenging setting, the observed gains demonstrate that multimodal knowledge distillation can substantially enhance event sensitivity and semantic understanding beyond near-random baseline performance.

From a systems perspective, this work highlights the effectiveness of the proposed **Implicit Perception** paradigm. Rather than explicitly computing optical flow or skeletal representations during inference, the student model internalizes multimodal motion cues through feature alignment with high-capacity teacher networks. As a result, the final model maintains low computational complexity while exhibiting perceptual capabilities that exceed those achievable from raw RGB supervision alone. Furthermore, the integration of the **Contextual Module (CTX)** plays a crucial role in practical deployment, leveraging the inherent temporal logic of tennis to correct visually ambiguous predictions and improve sequence-level reliability.

Beyond quantitative performance, this study demonstrates the broader applicability of the MD-FED framework for educational and grassroots sports settings. By reducing dependence on large-scale annotations and specialized hardware, the proposed system lowers the barrier to deploying advanced sports analytics on campus. This enables automated technical statistics, structured video indexing, and objective performance feedback

for players and coaches who previously lacked access to professional analytical tools.

Despite these contributions, several limitations remain. The student model’s performance is bounded by the quality of the teacher networks, and negative transfer may occur when the professional domain diverges significantly from campus conditions. In addition, extreme camera viewpoints and heavy environmental interference continue to challenge localization accuracy. These limitations point to promising future directions, including test-time adaptation, viewpoint-invariant representation learning, and the integration of explicit ball trajectory modeling.

In conclusion, this work provides compelling evidence that **multimodal knowledge distillation offers a scalable and practical pathway for bridging the gap between elite sports analytics and noisy, real-world educational environments**. By shifting from explicit multimodal inference to efficient implicit perception, the proposed framework lays a solid foundation for the next generation of accessible, interpretable, and deployable sports intelligence systems.

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